

Homework 17:

1.

```
/**
 * Reverses ("flips") {@code this}.
 *
 * @updates this
 * @ensures this = rev(#this)
 */
public void flip() {
    if( !this.isEmpty()){
        T front = this.dequeue();
        This.flip();
        This.enqueue(front);
    }
}
```

2.

```
/**
 * Reverses ("flips") {@code this}.
 * @param q
 * @updates q
 * @ensures q = rev(#q)
 */
public static <T> void flip(Queue<T> q) {
    if( !q.isEmpty()){
        T front = q.dequeue();
        Filp(q);
        q.enqueue(front);
    }
}
```

3.

- TATACAT
- AGCTGTTTTTCGTT
- CACTCCATTTTA
- CATTTTAGCTGTT
- **TTTCGTTA**TACAT
- CTGT**TTTCGTTA**

→

- **TATACAT**
- AGCTGTTTTTCGTT
- CACTCCATTTTA
- CATTTTAGCTGTT
- CTGTTTTTCGT**TATACAT**

→

- AGCTGTTTTCGTT
- CACTCCATTTTA
- CATTTTAGCTGTT
- CTGTTTTTCGTTATACAT

→

- CACTCCATTTTA
- CATTTTAGCTGTTTTTCGTT
- CTGTTTTTCGTTATACAT

→

- CACTCCATTTTA
- CATTTTAGCTGTTTTTCGTTATACAT

→

- CACTCCATTTTAGCTGTTTTTCGTTATACAT